

CSE-251

ELECTRONIC CIRCUITS

LECTURE-08

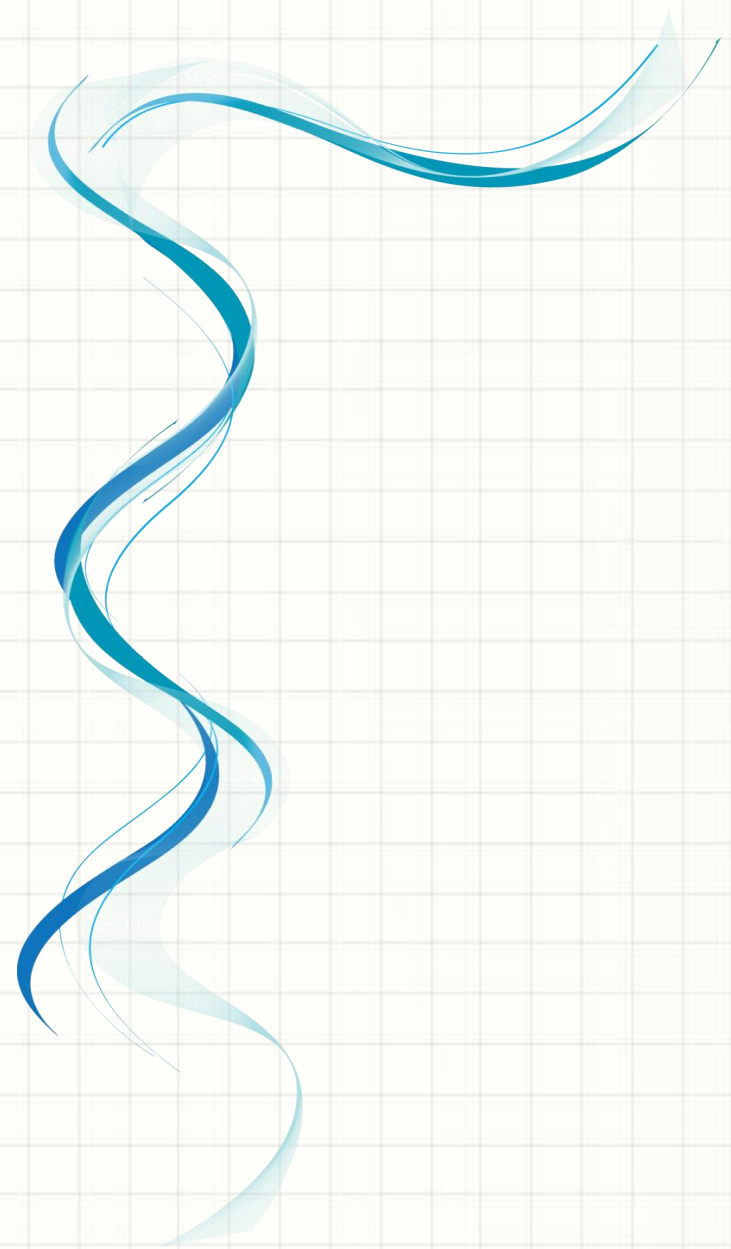
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Today's Overview:

- ❑ *Full-wave Rectifier*
- ❑ *Zener Diode*



Full-wave Rectifier



Full-wave Rectifier:

The process of rectification in which both half-cycles of the input become unidirectional in the output is called full-wave rectification.

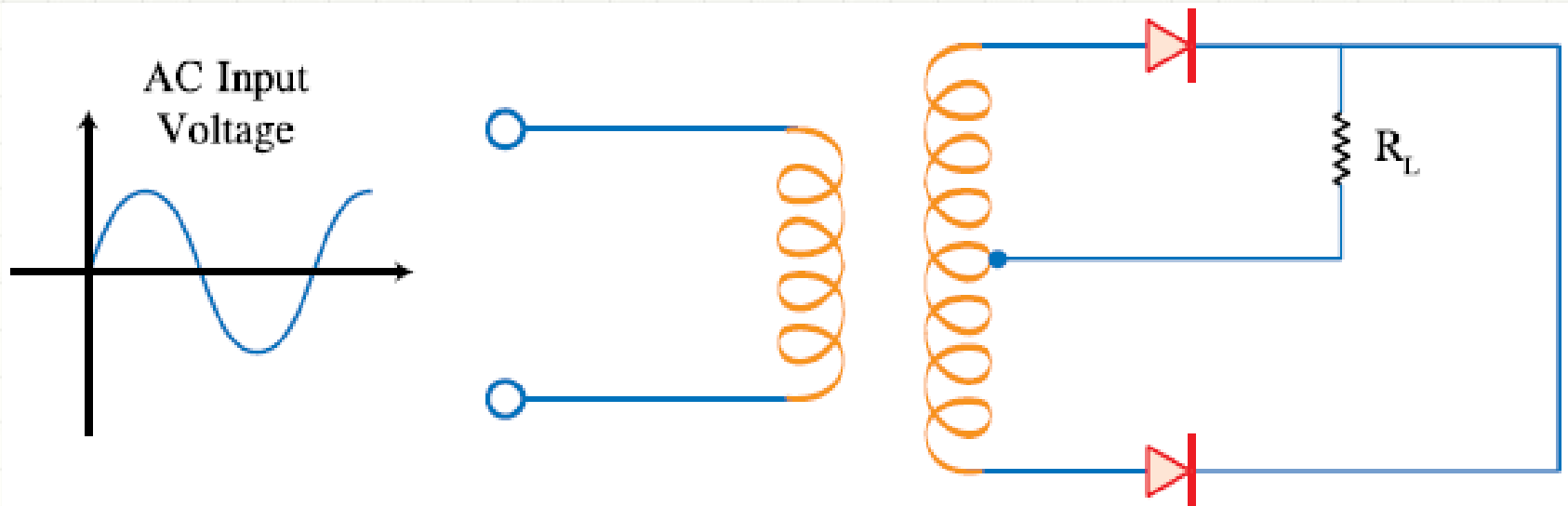
The **Full-wave rectifier** utilizes both halves of each ac input. Basically, there are two ways to perform full-wave rectification:

- 1) Center Tap / Two diode method.
- 2) Bridge Rectifier / Four diode method.

Full-wave Rectifier:

Center Tap Method / Two diode method:

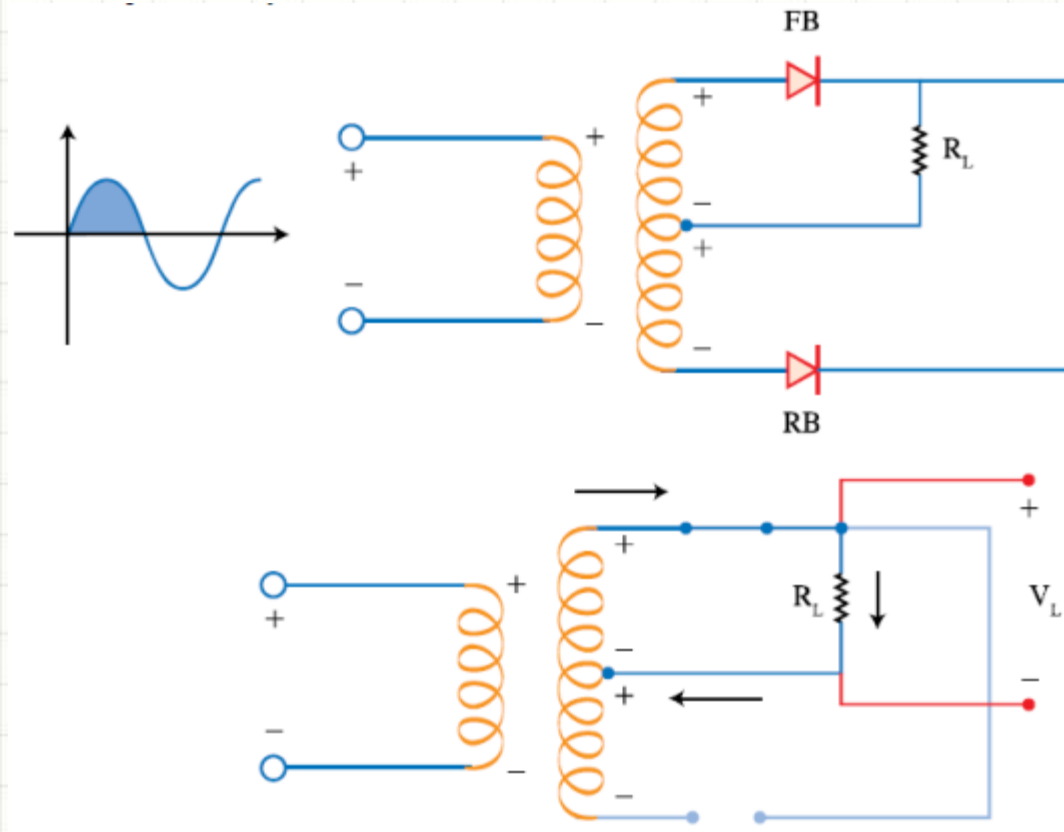
In this method, two diodes and a transformer are required for rectification. Another wire is connected to the center of the secondary end of the transformer through a load resistor so that the potential difference is divided into two equal parts. By connecting the two diodes as shown in the figure, rectification is possible easily.



Full-wave Rectifier:

Center Tap Method / Two diode method:

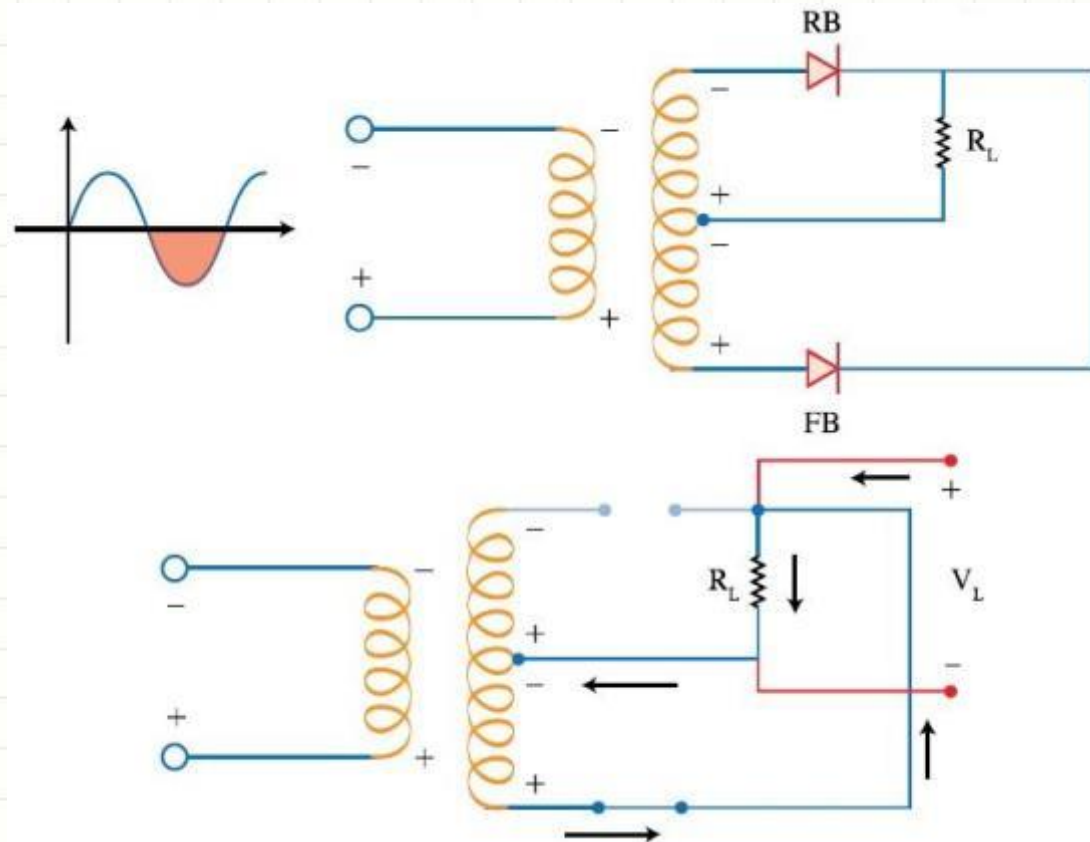
In the positive half-cycle, D_1 will be in forward bias, but D_2 will be in reverse bias. So, no current will flow through D_2 . The current in the secondary coil will flow from D_1 to the resistor R_L , from top to bottom (clockwise). If we assume that the two diodes are ideal diodes, then D_1 will be short-circuited in the forward bias. As a result, current will flow through the resistor in that line, and the output will be similar to the input voltage.



Full-wave Rectifier:

Center Tap Method / Two diode method:

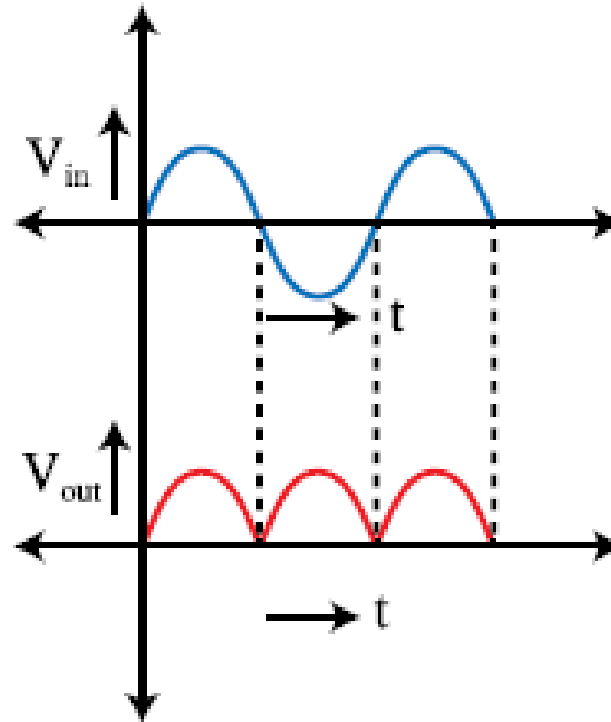
Again, in the negative half-cycle, the positive and negative ends of the line voltage will exchange places. As a result, D_1 diode will now be reverse biased, so it will not carry any current. D_2 diode will now be in forward bias, so it will be short circuited. But the direction of current flow in R_L resistor is the same. As a result, the voltage is also the same as in the positive half-cycle.



Full-wave Rectifier:

Center Tap Method / Two diode method:

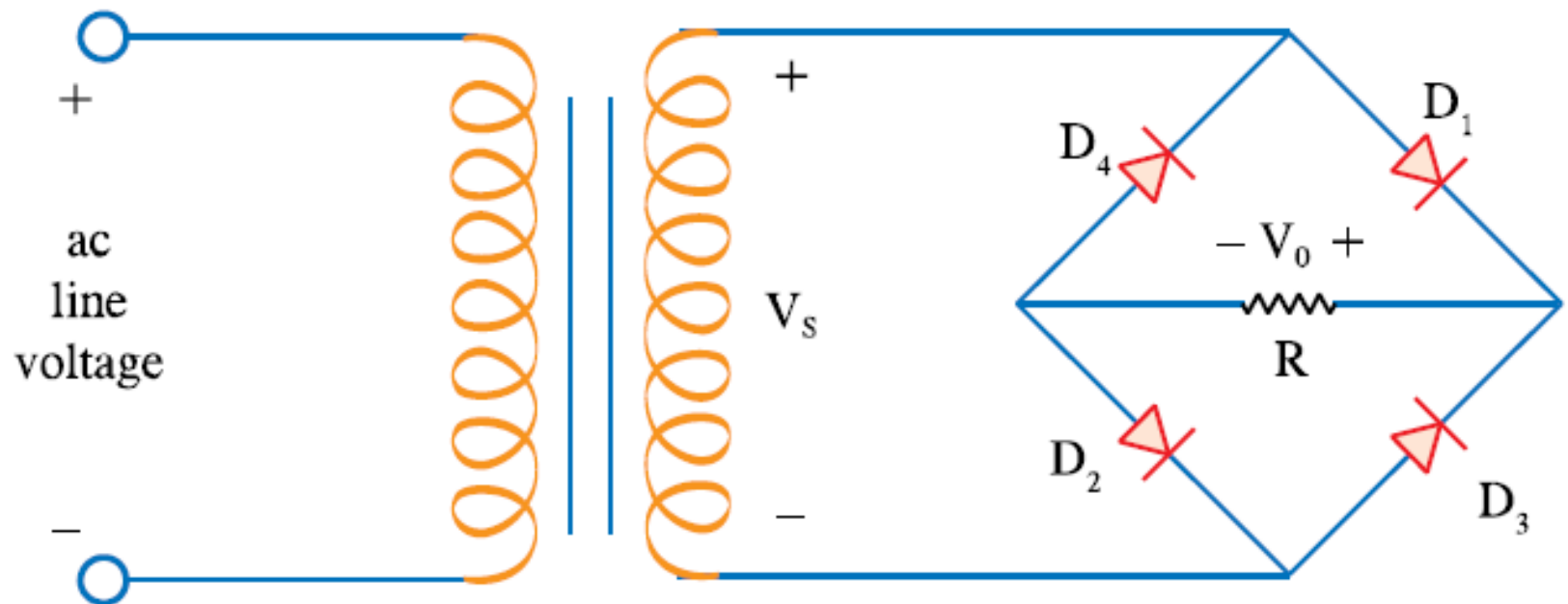
The current flows from top to bottom through the resistor in both half-cycles. As a result, if we measure the voltage difference between the two ends of the resistor, we will get a positive output in both half-cycles. In this way, the input has both positive and negative half-cycles, but the output has only positive half-cycles, and both half-cycles are used in this case. Therefore, full-wave rectification is achieved.



Full-wave Rectifier:

Bridge Rectifier / Four diode method:

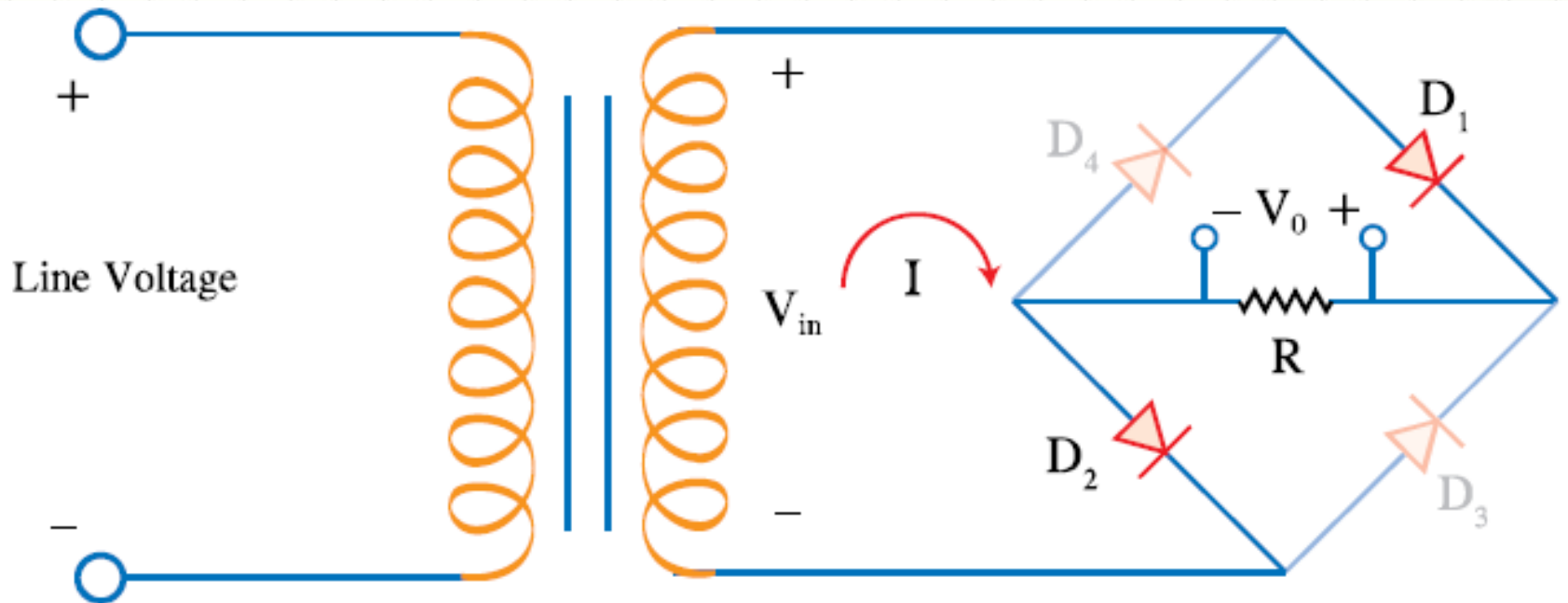
It is possible to achieve full-wave rectification without a center tap. However, this will require four diodes. These four diodes need to be connected according to the following figure. This connection is similar to a Wheatstone bridge, so it is called the bridge rectification method.



Full-wave Rectifier:

Bridge Rectifier / Four diode method:

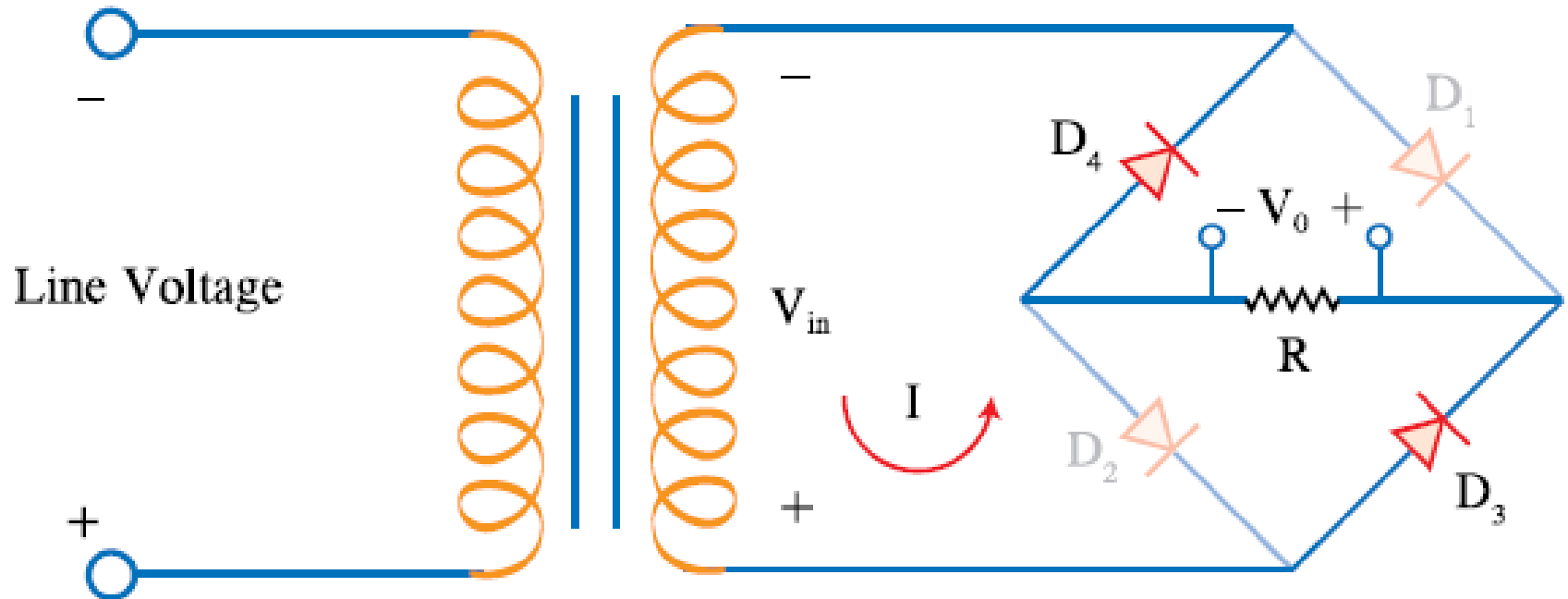
In the positive half-cycle, the positive and negative ends of the line voltage cause the D_1 and D_2 diodes to be in forward bias. The diodes D_3 and D_4 are in reverse bias for the same reason. The current flows through the diode D_1 to the resistor, and then through the D_2 diode to the negative end.



Full-wave Rectifier:

Bridge Rectifier / Four diode method:

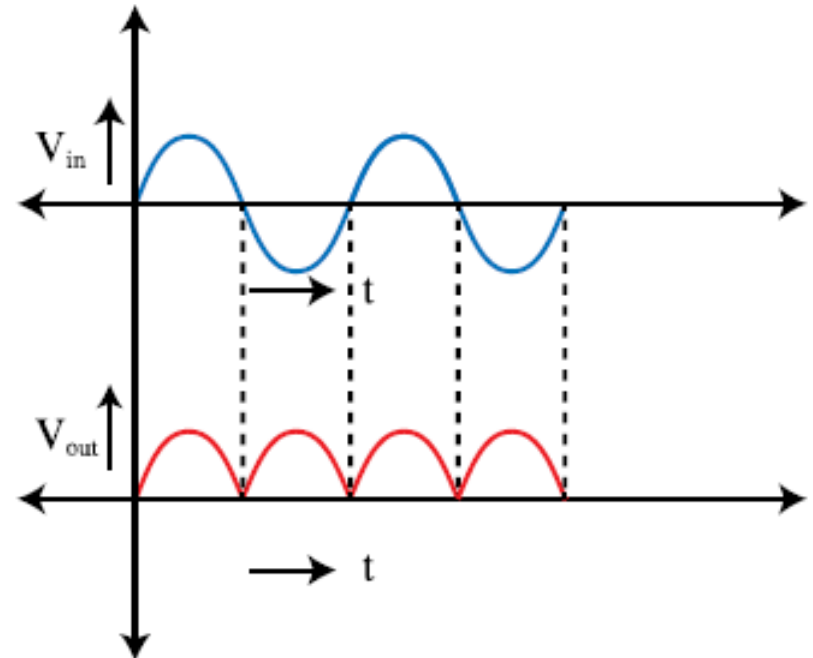
In the negative half-cycle, the positive and negative ends of the line voltage exchange places. As a result, the diodes D_1 and D_2 will now be in reverse bias, and the diodes D_3 and D_4 will be in forward bias.



Full-wave Rectifier:

Bridge Rectifier / Four diode method:

Therefore, the current flow through the resistor is in the same direction in both the positive and negative half-cycles. If we measure the voltage difference between the two ends of the resistor, the output voltage will always be positive. Since the output voltage is positive in both the positive and negative half-cycles, the full wave is rectified.



Full-wave Rectifier:

Average Value for full time period:

$$V = V_m \sin \omega t$$
$$\bar{V} = V_{av} = \frac{2V_m}{\pi} = 0.636 V_m$$

Full-wave Rectifier:

RMS Value for full time period:

$$V = V_m \sin \omega t$$
$$V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707 V_m$$

Full-wave Rectifier:

Form Factor:

Form Factor of a Full Wave Rectifier (F.F)

The form factor is the ratio between **RMS value** and average value.

$$FF = \frac{RMS\ value}{Average\ value}$$

The formula for form factor is given below:

The form factor of a centre-tapped full wave rectifier is equal to 1.11 (i.e. $FF = 1.11$).

Full-wave Rectifier:

Efficiency:

Rectifier efficiency (η) is the ratio between the output DC power and the input AC power.

The formula for the efficiency is equal to:

$$\eta = \frac{P_{dc}}{P_{ac}}$$

The efficiency of a centre-tapped full-wave rectifier is equal to 81.2% (i.e. $\eta_{\max} = 81.2\%$).

Full-wave Rectifier:

Advantages of Full Wave Rectifier:

- The rectifier efficiency of a full-wave rectifier is high
- The power loss is very low
- Number of ripples generated are less

Disadvantages of Full Wave Rectifier:

- Expensive

Full-wave Rectifier:

Applications of Full Wave Rectifier:

- Full-wave rectifiers are used for supplying polarized voltage in welding.
- Full-wave rectifiers are used for detecting the amplitude of modulated radio signals.

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Problem:

Determine the output waveform for the given network and calculate the output dc level.

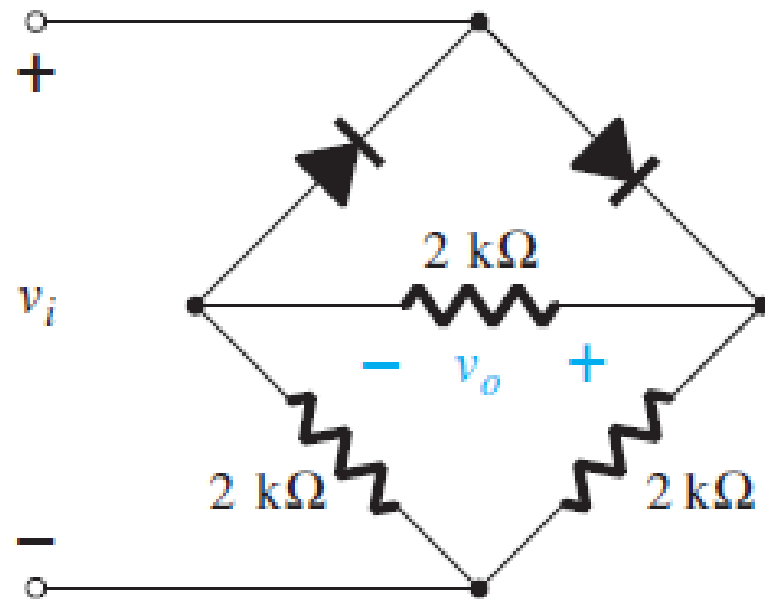
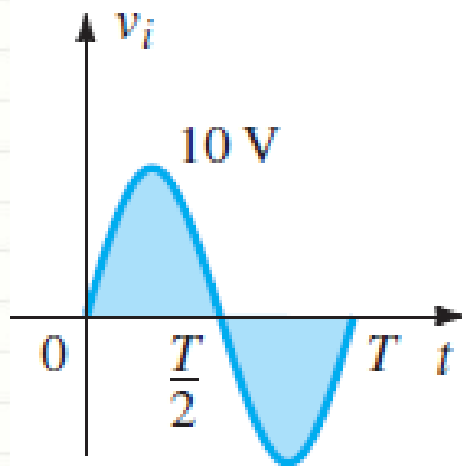


FIG. 2.64

Bridge network for Example 2.17.

Problem:

Determine the output waveform for the given network and calculate the output dc level.

Solution: The network appears as shown in Fig. 2.65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 2.66, where $v_o = \frac{1}{2}v_i$ or $V_{o_{\max}} = \frac{1}{2}V_{i_{\max}} = \frac{1}{2}(10 \text{ V}) = 5 \text{ V}$, as shown in Fig. 2.66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 2.67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{dc} = 0.636(5 \text{ V}) = 3.18 \text{ V}$$

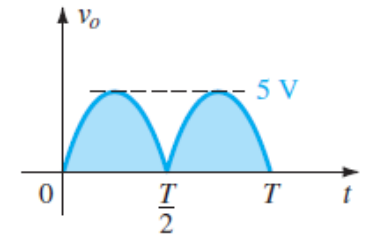


FIG. 2.67

Resulting output for Example 2.17.

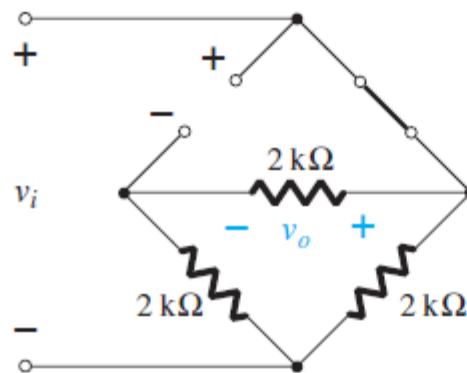
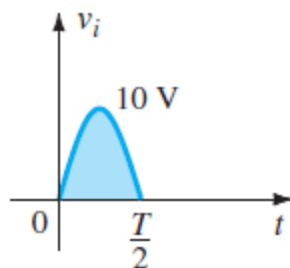


FIG. 2.65

Network of Fig. 2.64 for the positive region of v_i .

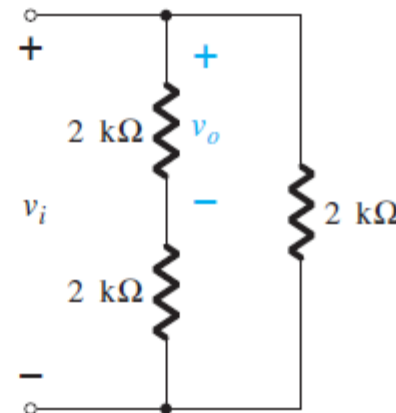
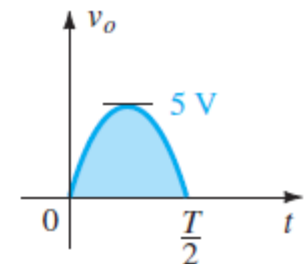
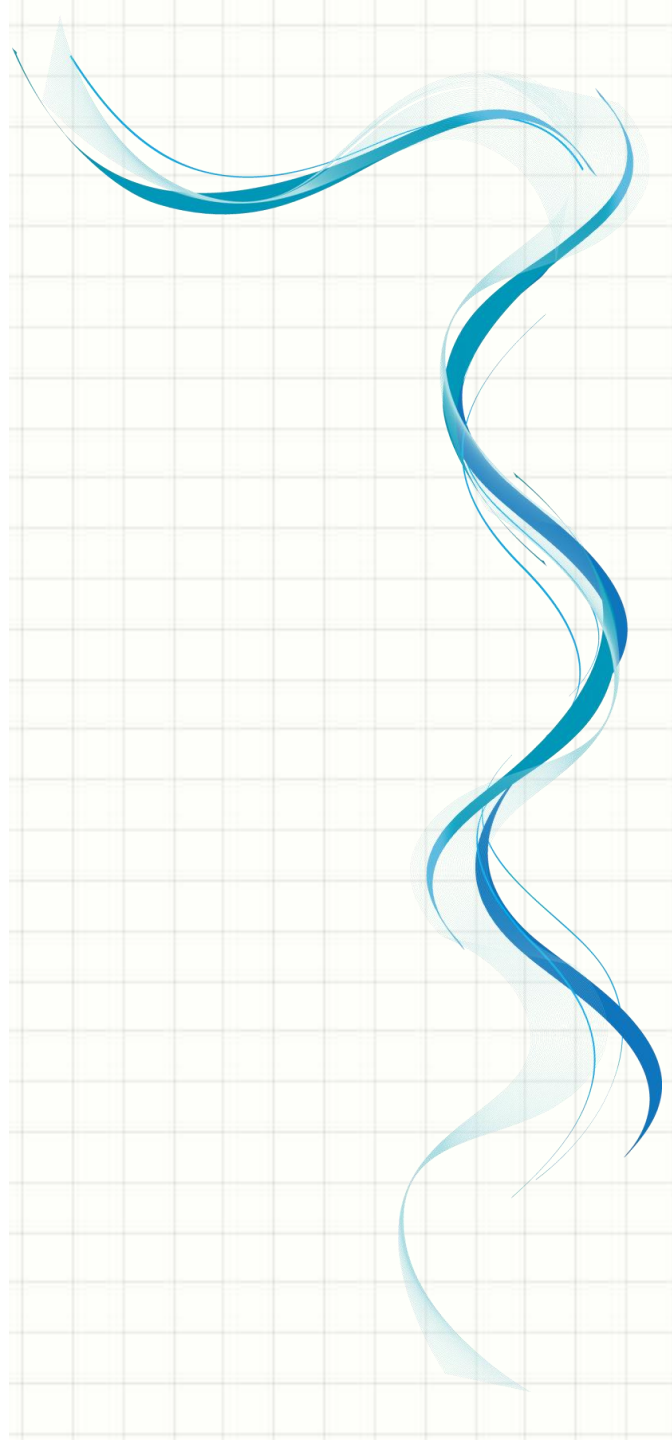


FIG. 2.66

Redrawn network of Fig. 2.65.



A decorative blue wavy line with a light blue shadow, flowing vertically on the left side of the slide.

Zener Diode

Zener Diode:

Clarence Melvin Zener was the first person to describe the electrical properties of Zener Diode.

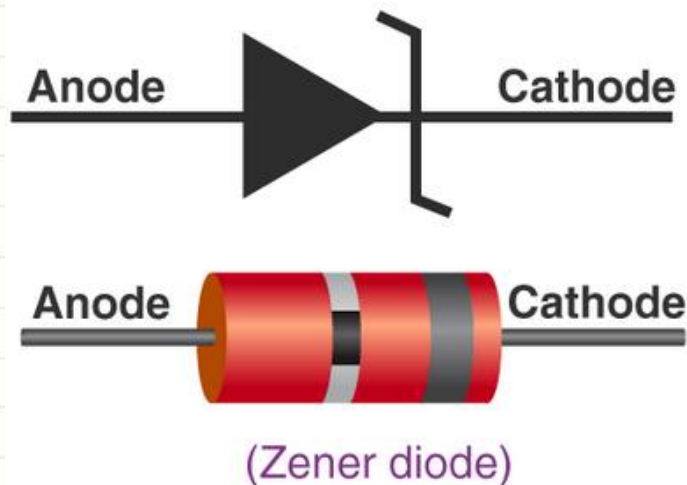
*A **Zener diode** is a heavily doped semiconductor device that is designed to operate in the reverse direction.*



Zener Diode:

The main difference between a **Zener diode** and a **normal diode** lies in the passage of current. A normal diode allows current to flow only in one direction while Zener diode allows current to flow in both directions.

A Zener diode operates just like a normal diode when it is forward-biased. However, a small leakage current flows through the diode when connected in reverse biased mode. As the reverse voltage increases to the predetermined breakdown voltage (V_z), current starts flowing through the diode.

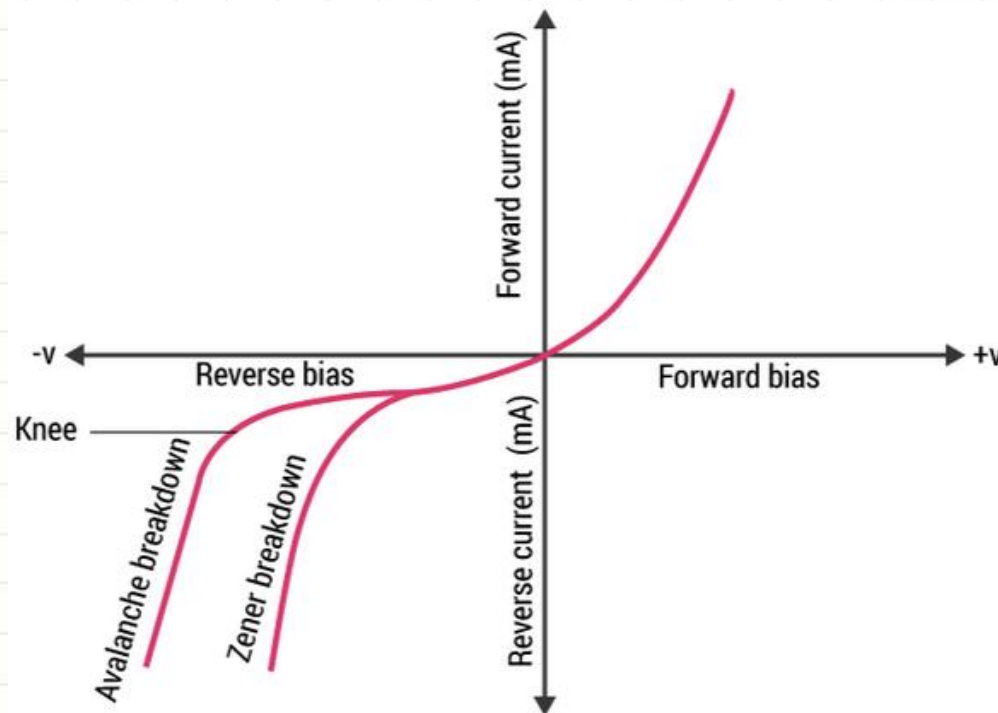


Zener Diode:

The I-V characteristics of a Zener diode can be divided into two parts as follows:

Forward Characteristics of Zener Diode:

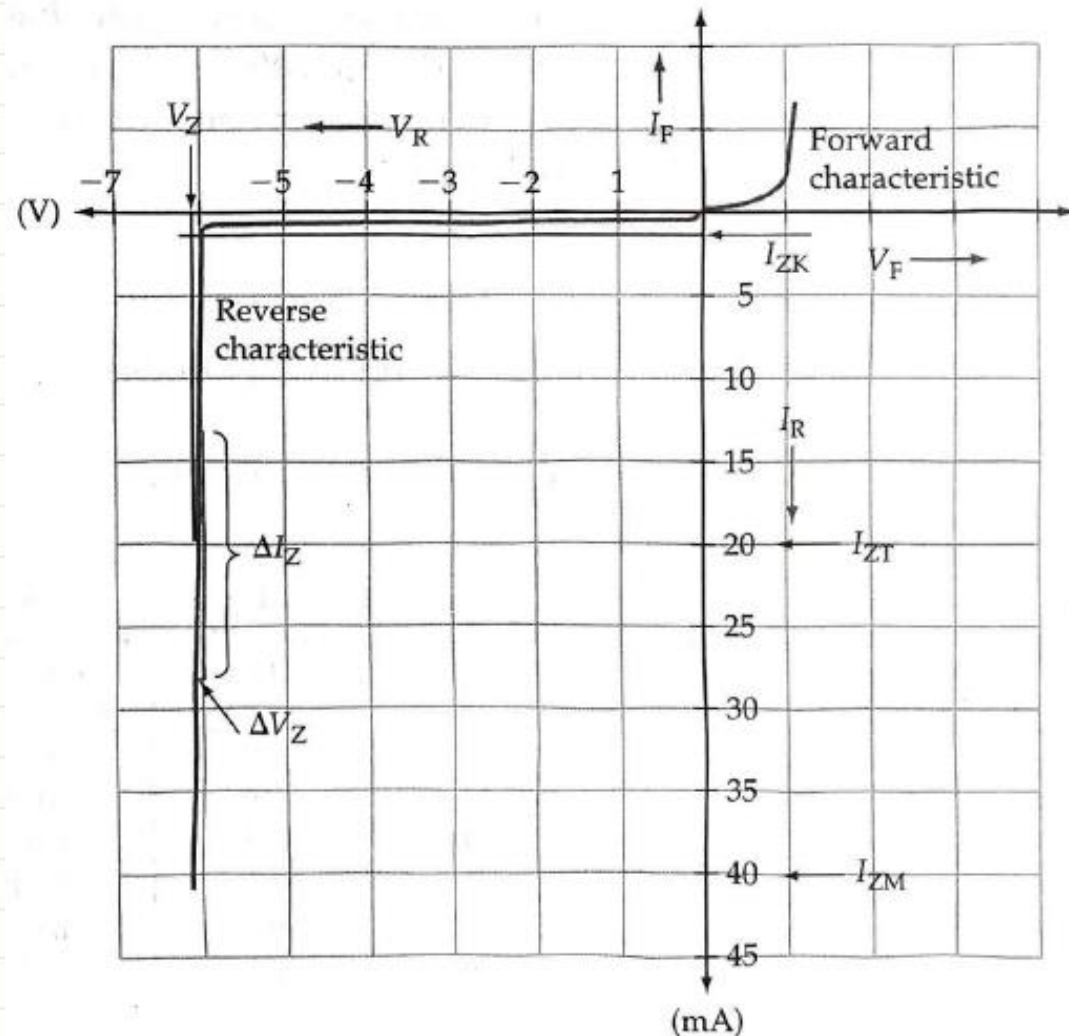
The first quadrant in the graph represents the forward characteristics of a Zener diode. From the graph, we understand that it is almost identical to the forward characteristics of any other P-N junction diode.



Zener Diode:

Reverse Characteristics of Zener Diode:

When a reverse voltage is applied to a Zener diode, a small reverse saturation current flows across the diode. As the reverse voltage increases, at a certain value of reverse voltage, the reverse current increases drastically and sharply. We call this voltage **Zener voltage**, and V_z denotes it.





QUESTIONS?

SEE YOU IN THE NEXT CLASS

